SMART EXPENSE TRACKER - USER MANUAL

Developed by **Disha Prajapati** | Complete Step-by-Step Guide to Deploy and Run on Any Laptop/PC

1. System Prerequisites

Before running this project on a new laptop, ensure the following minimum software environment is installed:

Prerequisite Component	Required Specification	Download / Verification Note
Python 3 Environment	Python 3.9, 3.10, 3.11, or 3.12	Official installer from python.org
Package Manager (pip)	Pip 22.0 or higher	Bundled automatically with standard Python installer
Operating System	Windows 10/11, macOS, or Linux	Fully cross-platform (No OS-specific dependencies)
Web Browser	Chrome, Edge, Firefox, or Safari	Required to view the Web Application Dashboard

CRITICAL STEP DURING PYTHON INSTALLATION: When installing Python on Windows, make sure to **CHECK** the box at the bottom that says: *'Add python.exe to PATH'*. If you skip this, terminal commands like 'python' or 'pip' will not be recognized.

2. Transferring and Extracting the Project

Step 1: Copy the complete project folder (SmartExpenseTracker) to a Pen Drive, Google Drive, or transfer as a ZIP archive.
Step 2: On the destination laptop, paste and extract the folder to any convenient location (e.g., D:\SmartExpenseTracker or C:\Users\YourName\Desktop\SmartExpenseTracker).

3. Installing Required Python Dependencies

Open Command Prompt (cmd) or PowerShell, navigate to the project directory, and install required libraries:

```

```

What does requirements.txt install?: It installs **FastAPI**, **Uvicorn** (for the web interface), **ReportLab** (for academic PDF generation), and **PyMuPDF**. Standard modules like sqlite3 and tkinter are already built into Python.

4. How to Run the Web Application (Website Interface)

The Web Application provides a modern browser-based interface where users can view charts, add transactions with live auto-categorization, and remove expenses with a single click.

Method A: Quick One-Click Double-Click (Easiest for Non-Technical Users)

1. Open the project folder in Windows File Explorer.
2. Double-click the file named **run_web.bat**.
3. A black console window will open, and your web browser will automatically open to **http://127.0.0.1:8000**!

Method B: Terminal / Command Prompt Command

Run the Python runner script from terminal:

How to Add & Remove Expenses on the Web App: • **To Add Expense:** Click **'+ Add Expense'** at the top. Type a description (e.g., 'Swiggy dinner', 'Uber cab', 'Electricity bill') — the system auto-detects the category! Enter amount and click **'Save'**.
• **To Remove Expense:** In the transactions table, find the expense and click the ■■ **(Trash Icon)**. Confirm the popup prompt, and the transaction is instantly deleted from SQLite database.

5. How to Run the Native Desktop Application (Tkinter GUI)

If you want to use the native desktop standalone software with tabs, canvas vector graphics, and rule-training:

6. Initializing Realistic Demo Data (Optional for Evaluation/Viva)

If you are demonstrating the project to an examiner, professor, or client and need realistic transactions, budgets, and charts pre-populated immediately, run:

7. Generating the Official 7-Page Academic Synopsis PDF

The project includes an automated ReportLab generator that produces a submission-ready 7-page academic synopsis PDF covering Introduction, Literature Review, Comparative Matrix, SDLC Model, Flowcharts, and Gantt Timeline:

The generated PDF is saved at: docs/Smart_Expense_Tracker_Synopsis.pdf.

8. Troubleshooting & Frequently Asked Questions (FAQ)

If you encounter any unexpected issues on a new laptop, refer to the diagnostic matrix below:

Error / Symptom	Root Cause	Immediate Solution & Fix
'python' is not recognized as an internal or external command	Python was installed without checking 'Add to PATH'.	Re-run the Python installer, select **'Modify'**, and check **'Add Python to PATH'**. Alternatively, search 'Environment Variables' and add Python install path.
ModuleNotFoundError: No module named 'fastapi' or 'reportlab'	Dependencies have not yet been installed in current Python environment.	Run command: pip install -r requirements.txt. Ensure your computer is connected to the internet during install.
Error: Port 8000 is already in use	Another process (or previous instance of web_app) is running on port 8000.	Open web_app.py and change port=8000 to port=8080 or port=5000, then restart.
Browser does not open automatically	Default browser handler delayed or restricted by Windows security.	Manually open Google Chrome, Microsoft Edge, or Firefox, and type **http://127.0.0.1:8000** in the address bar.
Where is transaction data stored?	Local storage persistence.	All data is stored locally in expense_tracker.db (SQLite). If you move the folder, all data stays intact with 100% privacy.

9. Project Quick Reference Summary Sheet

Feature / Objective	Terminal Command	Expected Output
Launch Web Interface	python run_web.py	Opens web dashboard at http://127.0.0.1:8000
Launch Desktop GUI	python app.py	Opens native Tkinter desktop application window
Populate Demo Data	python sample_data.py	Seeds realistic transactions, budgets & categories
Generate Synopsis PDF	python generate_synopsis_pdf.py	Generates 7-Page Academic Synopsis PDF